

CURRICULUM VITÆ

PERSONAL INFORMATION

Tanmoy Sil
+49 152 220 66828
tanmoy_sil@hotmail.com
LinkedIn · GitHub · Google Scholar

SUMMARY

Biomedical engineer and ML researcher with a strong publication record in clinical AI and signal processing for Parkinson's disease. Combines deep learning and neural signal analysis with hands-on systems engineering, from closed-loop deep brain stimulation biomarkers to clinical-grade motion capture laboratories, to deliver validated diagnostic and therapeutic tools.

EDUCATION

2021–2026	PhD, Computational Neuroscience <i>University of Würzburg, Germany</i> Completed: 2026 · Grade: 1.0 (magna cum laude)
2018–2019	MS, Biomedical Engineering <i>University of Southern California, Los Angeles, USA</i> Completed: December 2019
2012–2016	BTech, Electronics & Instrumentation Engineering <i>West Bengal University of Technology, Kolkata, India</i> Completed: July 2016

RESEARCH EXPERIENCE

2026–	Research Associate <i>University Clinic Würzburg, Germany</i> - Conceived and delivered a permanent clinical-grade 8-camera 3D human movement analysis laboratory from the ground up: defined system specifications, led procurement, designed optical layout, fabricated custom electronics, and built a pose estimation and 3D triangulation pipeline for quantitative gait and spasticity assessment - Specified, procured, and built a synchronized multi-camera system for biomechanics and muscle movement research in cervical dystonia, with hardware triggering, real-time acquisition, a PyTorch-based 3D triangulation pipeline, and video archival
2021–2026	PhD Candidate <i>University Clinic Würzburg, Germany</i> - Developed robust framework for identifying beta bursts in Parkinson's disease LFP using wavelet decomposition, analyzing correlations with motor impairment for closed-loop DBS applications - Created classification model differentiating Parkinson's disease from Essential tremor using peripheral signals (EMG, accelerometer), achieving substantial improvements over published metrics

- Translated rat model findings to humans, examining beta burst amplitude relationships between STN LFP and motor cortex, with striatal dopamine as mediator
- Applied a Generative Causal Explanations (GCE) framework to STN local field potentials in DBS patients, separating stimulation presence from therapeutic response in latent space to identify a stimulation-orthogonal biomarker of motor improvement
- Derived an interpretable latent motor biomarker (Kinematic Discriminative Component) from markerless video kinematics, tracking medication and DBS effects more sensitively than clinician-rated MDS-UPDRS scores; validated prospectively in two independent cohorts

2015–2016

Bachelor's Thesis Research

West Bengal University of

Technology, India

- Designed data-driven control system for DC motor positioning using MATLAB system identification and PID control with repetitive strategies

PUBLICATIONS

Perales, M.†, **Sil, T.†**, et al. (2026). An interpretable latent motor axis tracks treatment and progression in Parkinson's disease. *npj Digital Medicine* [To be submitted].

Sil, T., Navratil, P., et al. (2026). A stimulation-orthogonal neural signature of therapeutic response in Parkinsonian deep brain stimulation. *Nature Biomedical Engineering* [In Submission].

Asadi, A., Peach, R., **Sil, T.**, et al. (2026). Parkinsonian beta bursts and deep brain stimulation: from subthalamic LFP electrophysiology to biophysical modelling. *Brain Stimulation* [In Submission].

Sil, T., et al. (2026). Sensor-based data-driven differentiation between Parkinson's tremor and essential tremor. *Expert Systems with Applications*, 299, 130336. <https://doi.org/10.1016/j.eswa.2025.130336>

Hofman, K.†, Chen, J.Z.†, **Sil, T.**, et al. (2025). Low β predicts motor output and cell degeneration in the A53T Parkinson's disease rat model. *Brain*, 2025: 00; 1–14. <https://doi.org/10.1093/brain/awaf063>

Sil, T., et al. (2023). Wavelet-based bracketing, time-frequency beta-burst detection: new insights in Parkinson's disease. *Neurotherapeutics*, 20, 1767–1778. <https://doi.org/10.1007/s13311-023-01447-4>

TEACHING & PROFESSIONAL EXPERIENCE

2024–2025

Course Instructor – Data Science *University Clinic Würzburg, Germany*

Delivered lectures and led coding tutorials for graduate-level data science courses during summer semesters.

2018–2019

Faculty Liaison *USC Rossier School of Education, Los Angeles, USA*

Coordinated between faculty and administration; managed admissions for 336 students.

2018–2019 **STEM Instructor** *USC Viterbi School of Engineering, Los Angeles, USA*
Mentored 50 primary school students through the USC Viterbi STEM Educational Outreach Program.

TECHNICAL SKILLS

Research	Signal Processing, Machine Learning, Deep Learning, Computer Vision, Pose Estimation, Camera Systems
Programming	MATLAB, Python (PyTorch, scikit-learn, OpenCV, SciPy, NumPy)
Management	Project Management, Data Analysis, Cross-functional Collaboration
Languages	Bengali (native), English (fluent), Hindi (fluent), German (beginner)

HONORS & AWARDS

2022	PhD Fellowship, Deutscher Akademischer Austauschdienst (DAAD), Germany
2020	Featured Graduate Student, USC Viterbi School of Engineering, USA
2015	Best Intern Project Award, Variable Energy Cyclotron Centre, India

REFERENCES

Available upon request.